

Emiliano de la Garza Villarreal

✉ emiliano.gramm@gmail.com ☎ (470) 358-5294 🌐 emiliano-gramm.github.io

Education

University of Tennessee Chattanooga , (M.S.) Computer Science and Artificial Intelligence	Aug 2026 – May 2028
University of Tennessee Chattanooga , (B.S.) Data Science	Aug 2024 – May 2026
Cleveland State Community College , (A.S) Computer Science	Aug 2022 – Dec 2023

Experience

CUIP , Machine Learning Researcher - Built a real-time intersection monitoring system on NVIDIA Jetson using ROS2 and CUDA to deploy a federated-trained model, implementing multi-processing to resolve CPU image-processing bottlenecks and ensure continuous GPU saturation for reliable, low-latency inference. - Achieved state-of-the-art trajectory prediction by designing a federated learning strategy within a Flowchain model, combining FedProx and FedAdam to improve ADE and FDE metrics by ~8% compared to baseline results. - Developed a custom YOLO crash detection model by gathering open-source data and applying transfer learning, then improved accuracy and recall using Mosaic and MixUp augmentation.	Apr 2023 – present 3 years 2 months
Eaton Corporation , Engineering Intern - Analyzed large-scale industrial datasets to identify operational trends and inefficiencies, translating raw metrics into actionable insights to generate over \$300,000 in annual savings. - Conducted shop-floor time studies to build a predictive production model, updating system standards that reduced labor hours by 20%.	Dec 2023 – present 2 years 6 months

Projects

Swarm-Sense (ROS 2 & Distributed Systems)

- Architected a high-performance ROS 2 Humble fleet interface to monitor 5+ robots simultaneously via Foxglove Bridge and server-client WebSockets.
- Reduced video latency to sub-100ms by implementing hardware-accelerated H.264 transport and 500kbps bitrate caps for bandwidth-constrained environments.
- Created a Docker environment to improve team collaboration, reliability, and ease of extension for the multi-robot telemetry dashboard.
- Improved project documentation with reproducible run steps to increase maintainability and onboarding speed.
- Implemented comprehensive error handling and unit tests across the stack to improve robustness and overall software quality.

Character-Level Sequence Modeling (LSTM Exploration)

- Implemented a two-layer PyTorch LSTM for next-character prediction using fixed 50-step context windows and one-hot encoded inputs.
- Built a full preprocessing and training pipeline on a cleaned Sherlock Holmes corpus, using shifted targets with Adam + CrossEntropy over 100 epochs.
- Solved key training bottlenecks by moving from overlapping to non-overlapping windows to reduce memory usage, then flattening sequence tensors to satisfy loss-shape requirements; studied RNN/GRU concepts in the same module for architectural context.

CNN Image Upscaling

- Built an encoder-decoder CNN in PyTorch to upscale grayscale images from 28x28 to 56x56 using Conv2d, MaxPool2d, and Upsample blocks.
- Trained for 100 epochs with BCE loss and Adam, reaching final losses of 0.5625 (train) and 0.5651 (test).
- Improved experimentation workflow by caching processed datasets as NumPy arrays and resolving tensor-shape formatting for stable batched training.

CUDA parallel vector addition

- Tutorial from *Programming Massively Parallel Processors*—element-wise float addition with `cudaMalloc / cudaMemcpy`, a bounds-checked `__global__` kernel, and CMake + `CUDA::cudart` ([repo](#)).

Skills

Machine Learning and Deep Learning: PyTorch, Computer Vision (YOLO), TensorRT, Federated Learning

Data Engineering and Processing: RTSP video pipelines, Time-series analysis, Pandas, NumPy, OpenCV

Programming and Development Tools: Python, Multiprocessing and threading, C++, CUDA, Java, Git, Linux

Languages

English

Fluent

Spanish

Fluent